Circular Motion Kinematics

Synopsis: I summarize the rotational kinematics equations. Note that rotational kinematics is just like linear kinematics. Instead of linear displacement, you have an angular displacement since the motion has circular symmetry, so the distance from the center of rotation is fixed, and only the angle subtended by the motion changes while rotating. There are equations for angular velocity and angular acceleration that look just like the equations for velocity and acceleration in linear kinematics.

The angular displacement gives the arc length or the actual distance covered, and the angular velocity gives the tangential velocity for a given radius of circular trajectory (see below). Angular displacement is θ, and the arc length is $s = r\theta$, provided θ is in radians. That proviso is why we bother with radians at all: in degrees the same statement needs a conversion factor bolted onto it, and you would carry that factor through every line of every derivation that follows. Angular velocity is $\omega = \Delta\theta/\Delta t$ in rad/s, and the tangential velocity it gives you at radius r is $v_t = r\omega$. Notice that ω has no r in it and v_t does: every point on a rigid turning object has the same angular velocity, and a different speed.

That is as far as this Note goes. Note 2 picks up the other half of the story – centripetal acceleration and the force that supplies it – and the later notes turn α back on.

Performance Expectations: Apply rotational kinematics equations to solve circular and centripetal force problems.

For PHY 151 — with calculus: Define the angular variables as derivatives, $\omega = d\theta/dt$ and $\alpha = d\omega/dt$, recover the kinematics equations by integration, and get $a = -\omega^2 r$ by differentiating the position vector twice.

Before you read on – predict. Two children sit on the same merry-go-round, one near the hub and one out at the rail. Are they moving at the same speed, and are they turning at the same rate? Commit to a sentence here, before you watch. You will come back to it at the end.

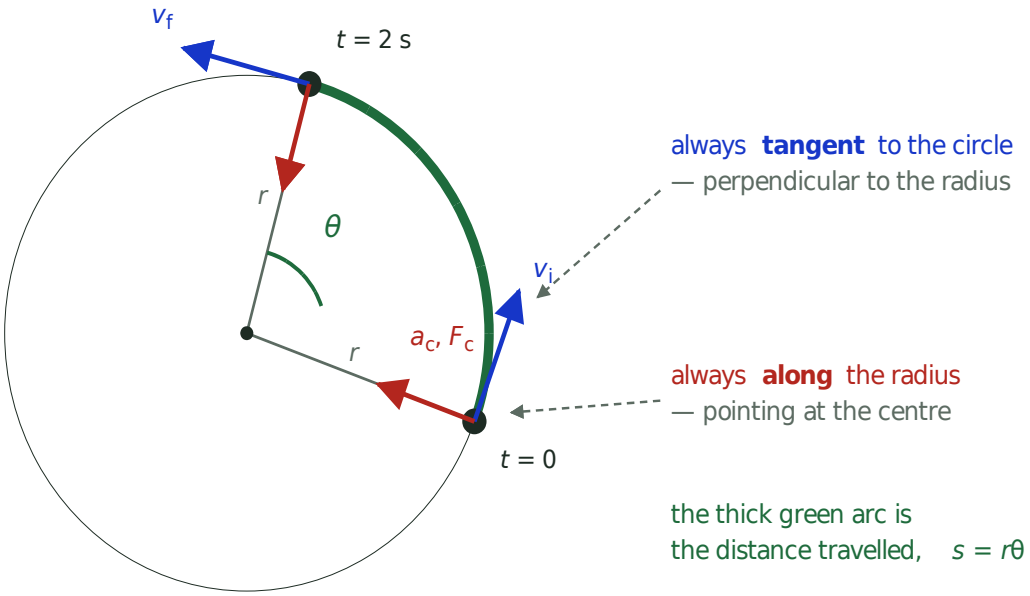
Same speed? Same rate? One sentence.

Instructions and Questions

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1) Angular displacement, and the radian.

Fix a center and a radius and the only thing left to say about a particle on that circle is its angle. Two things to keep straight: the arc is the distance actually travelled, so s = rθ is a real distance in metres; and that formula is true in radians and false in degrees.



Fill in the four names for one full turn, and the arc-length formula with the condition that makes it true:

1 _____ = 1 _____ = 1 _____ = _____° = _____ radians
arc length s = _____ true only when θ is in _____
degrees → radians: × _____ radians → degrees: × _____ π rad = _____ °

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Complete the table.

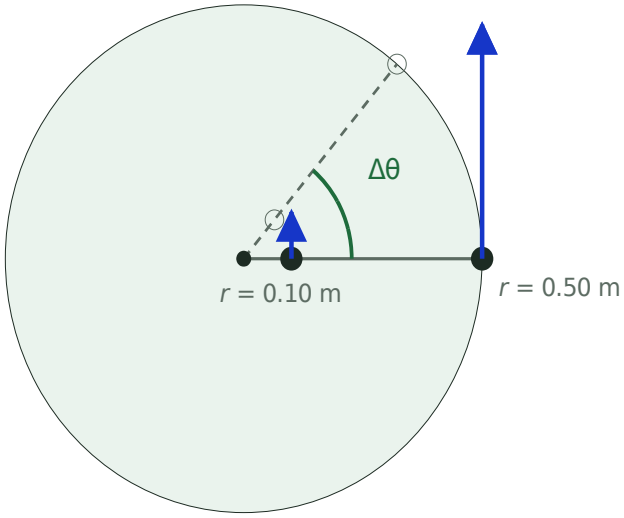
Angle	**in radians**	**in degrees**	**in revolutions**
10°		—	
50°		—	
75°		—	
260°		—	
π/3 rad	—		
0.55π rad	—		
1.8π rad	—		
4π rad	—		

i) A sector of radius 2.0 m subtends 75°. Find the arc length twice – once as $d = 2\pi r(\theta°/360)$, once as $s = r\theta$ – and say why they agree.

2) Angular velocity, and the tangential speed it gives you.
Every rotating machine in every problem is quoted in rev/min, and every equation below wants rad/s, so the first line of the working is almost always the same conversion. Write the three you need:

ω = _____ units: _____ v_t = _____
rev/min → rad/s: ω = (rev/min) × _____ ≈ (rev/min) × _____

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Both riders sit on the same spoke of the same merry-go-round.

Δθ is the angle the spoke sweeps; the open circles are where each rider has got to after that same interval.

Two riders on one spoke of the same merry-go-round. Radii and velocity arrows are both drawn to scale.

i) Microwave ovens rotate at a rate of about 7 rev/min. (A) What is this in revolutions per second? (B) What is the angular velocity in radians per second?

ii) A merry-go-round makes 2.5 rotations in 20 s. (A) What is its angular velocity? (B) What is the speed of a rider sitting 0.50 m from the center? (C) And of a rider sitting 0.10 m from the center?

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iii) In lacrosse, a ball is thrown from a net on the end of a stick by rotating the stick and forearm about the elbow. If the angular velocity of the ball about the elbow joint is 30.0 rad/s and the ball is 1.2 m from the elbow joint, what is the velocity of the ball?

iv) From your answers to the merry-go-round: which of ω and v_t depends on where you sit, and which does not? One sentence.

3) The rotational kinematics equations.
Define α = Δω/Δt, the angular acceleration, in rad/s^2. Now every constant-acceleration equation from Module 1 has a twin with θ in place of d – not an analogy, the same derivation in different letters. If you know the left column you already know the right one. This is the page you will turn back to for the rest of the module, so make it the neat one.

Linear Kinematics	**Rotational kinematics**
$d = d_0 + vt + \frac{1}{2}at^2$	
$v = v_0 + at$	
$v^2 = v_0^2 + 2ad$	
arc length	
tangential velocity	
tangential acceleration	
centripetal acceleration	
centripetal force	

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i) One of the bridge rows plays no part anywhere in this module. Which one, and why not?

ii) What unit must ω be in for every row to work, and by what factor is an answer wrong if you feed in rev/min instead?

4) For PHY 151 — with calculus.
The ratios above are averages over Δt. The instantaneous quantities are the derivatives, exactly as in linear kinematics: $\omega = d\theta/dt$ and $\alpha = d\omega/dt = d^2\theta/dt^2$. Written that way, the three rotational kinematics equations are not an analogy you have to trust – they are what you get by integrating α twice, which is the Module 1 calculation in different letters.

i) Integrate α once and then twice to recover $\omega = \omega_0 + \alpha t$ and $\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$, and say exactly where the assumption "α constant" is used.

ii) For uniform circular motion the position vector is $r(t) = r(\cos \omega t)\, i + r(\sin \omega t)\, j$. Differentiate once and verify two things about the velocity: that it is perpendicular to r at every instant, and that its magnitude is rω. You have just proved the two statements the first figure asserted.

Back to your prediction. Go back to the box on the first page. Was it the same rate, the same speed, both, or neither? Correct it here in one sentence, using ω and v_t by name.

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Check yourself

Four to do once you have filled the notes in. Work them here, then check them against the interactive version at the bottom of the note page.

1) A wheel turns at 45.0 rev/min. What is its angular velocity in rad/s?

2) Two riders sit on the same turning merry-go-round, one at 1.0 m from the hub and one at 3.0 m. Which statement is true?
   i) Same tangential speed, and the outer rider has one third the angular velocity
   ii) Same angular velocity, and the outer rider has three times the tangential speed
   iii) Same angular velocity and the same tangential speed – it is one rigid object
   iv) The outer rider has three times the angular velocity and three times the tangential speed

   Why?

3) The arc length formula $s = r\theta$ is
   i) true for any unit of angle
   ii) true only when θ is in radians
   iii) true only when θ is in degrees
   iv) true only for a whole number of revolutions

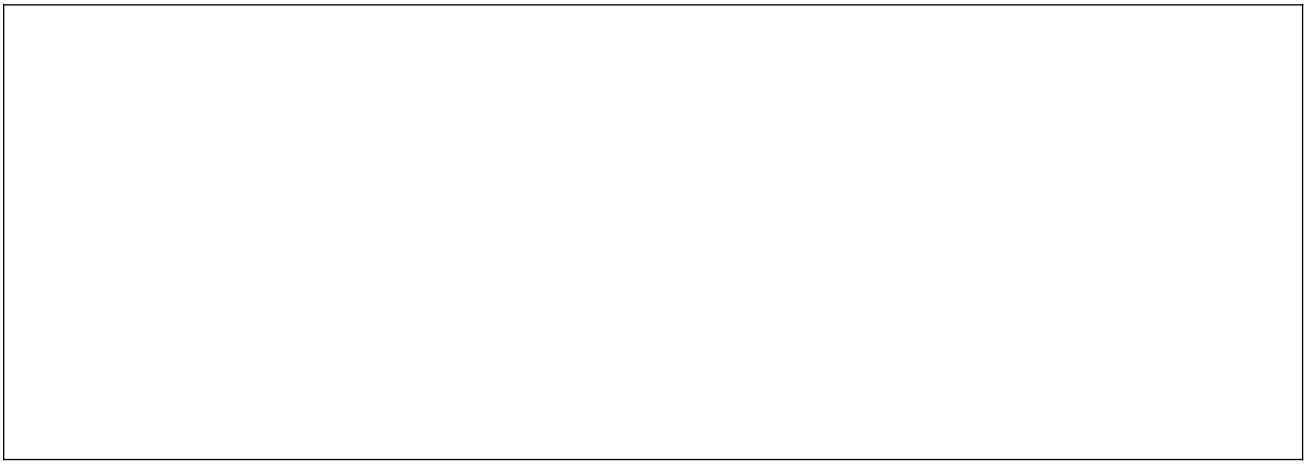
   Why?

4) A wheel of radius 0.30 m turns through 2.0 rad. How far does a point on its rim travel, in m?

Your notes

Space for the sketches, the derivations you want to redo yourself, and anything from the video that did not fit above. Everything on this handout is uniform circular motion; the later notes in this module turn α back on and take the same angular variables into rotational dynamics – torque, moment of inertia and angular momentum.

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